

WASP-12b

R-1975

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151121 · created 2026-08-07T00:17:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457347.76056 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.044
Transit depth (Rp/Rs)^2	0.00017 +/- 0.012 [%]
Orbital Inclination (inc)	85.46 +/- 3.0
Airmass coefficient 1 (a1)	1.52 +/- 0.4
Airmass coefficient 2 (a2)	-0.275 +/- 0.051
Scatter in the residuals of the lightcurve fit is	25.0 %
Best Comparison Star	None
Optimal Aperture	4.91
Optimal Annulus	5.73
Transit Duration (day)	0.1159 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.044 vs 0.1178 (deviation 2.65σ)
Duration fit vs published	0.1159 d vs 0.125 d (deviation 1.57σ)
Mid-time O–C	0.3 min
Depth SNR	0.0
Residual scatter	25.0 %

Light curve

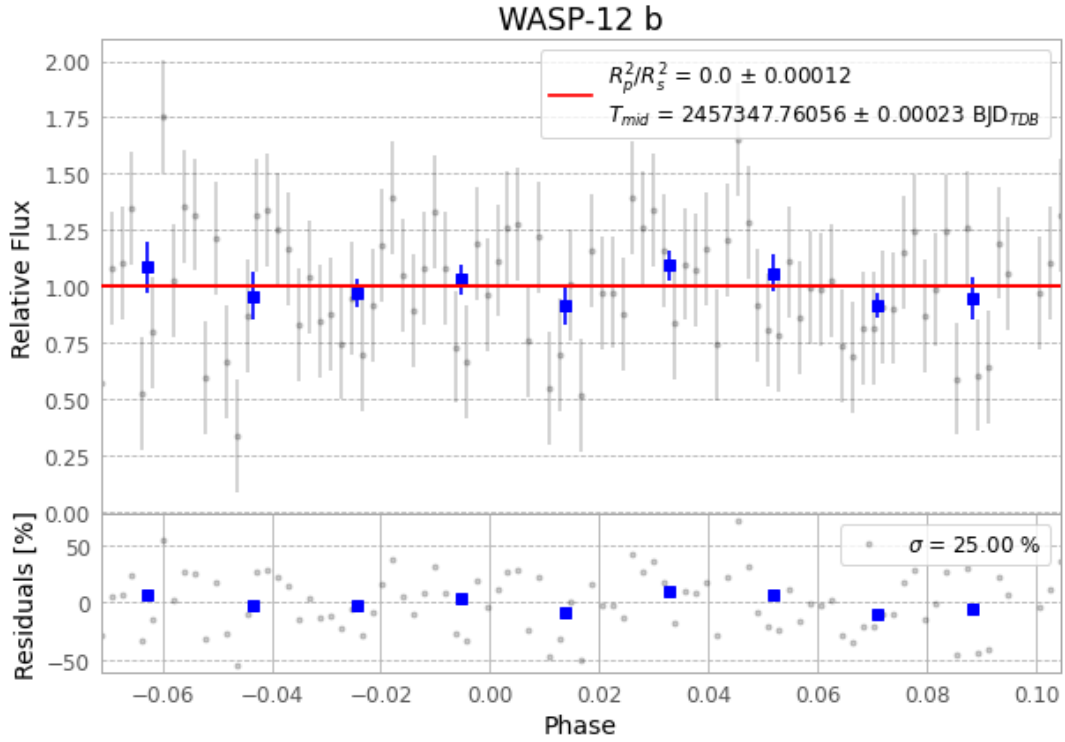


Figure 1 — FinalLightCurve_WASP-12 b_2015-11-20.png (SHA-256 5efc69eb6899e373...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [56, 383], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f24d85aab133e059...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

91 FITS frames (60 MB), WASP-12151121041512.FITS → WASP-12151121085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151121.log (d33a3e71c050...), FinalLightCurve_WASP-12 b_2015-11-20.png (5efc69eb6899...), FinalLightCurve_WASP-12 b_2015-11-20.csv (3ce1659c75e2...)

Compute resources

Wall time 127.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:17Z.